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RESEARCH ARTICLE

Bench-CoE: A Framework for Collaboration of Experts from Benchmark

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Abstract

Large Language Models (LLMs) are key technologies that drive intelligent systems to handle multiple tasks. To meet the demands of various tasks, an increasing number of LLMs-driven experts with diverse capabilities have been developed, spreading from language to visual understanding and generalization, accompanied by corresponding benchmarks to evaluate their performance. This paper proposes the Bench-CoE framework, which enables Collaboration of Experts (CoE) by effectively leveraging benchmark evaluations to achieve optimal performance across various tasks. Bench-CoE consists of a set of specialized expert models, a router for assigning tasks to corresponding experts, and a benchmark dataset for training the router. Based on this framework, we first formulate Query-Level Bench-CoE that is an abstraction of existing CoE methods exploiting the benchmark dataset. We further propose Subject-Level Bench-CoE, a new method that effectively addresses the potential issues of Query-Level Bench-CoE in poor generalization and labeling costs during training the router. Experiments show that the Query-Level Bench-CoE excels in in-distribution tasks, while the Subject-Level Bench-CoE demonstrates stronger out-of-distribution generalization and cross-domain scenarios adaptability. The codes are available at: https://anonymous.4open.science/r/BenchCoE.

Keywords

Large language models; Large multimodal models; Collaboration of experts; Expert routing; Benchmark evaluation; Out-of-distribution generalization

1 Introduction

Large Language Models (LLMs) are capable of performing various natural language processing (NLP) tasks through auto-regressive prediction conditioned on task prompts [1, 2]. The ability of LLMs to describe and unify tasks makes them key components in current visual understanding tasks, leading to the emergence of Large Multimodal Models (LMMs) [3, 4]. While these LLMs/LMMs-driven experts can handle a wide range of visual and language tasks, they possess different areas of expertise and exhibit significant performance variations across different tasks. As the capabilities of experts continue to improve, benchmark evaluations have also become increasingly complex and diverse [5–7]. For instance, benchmark such as MMLU [8] is used to assess multi-subject reasoning abilities in language tasks, while MMMU [9] evaluates cross-domain reasoning in multimodal tasks. It is difficult for a single model to achieve optimal performance across all tasks. Moreover, attaining comparable effectiveness often requires an increase in model scale and inference cost, which in turn poses challenges for computational resources and practical applications.

To integrate the advantages of the model in various aspects while controlling costs, researchers have proposed different approaches. Mixture of Experts (MoE) [10–12] introduces multiple expert models and adopts a sparse activation strategy, where only a subset of experts is activated during inference, thereby reducing computational cost. However, the experts in MoE is just sub-modules of a model, and it is difficult in understanding the functionality of each expert and the decision-making process in solving certain tasks. To mitigate this issue, researchers proposed Collaboration of Experts (CoE) [13–16], which integrates the advantages of different models by routing query to specific expert model, typically implemented through an expert router. However, these CoE methods rely on additional labeled data to train the router, and their generalization ability is often limited when handling out-of-distribution (OOD) tasks, posing challenges for expanding the capabilities of CoE models. Is there a way to maintain generalization without the additional cost of labeled data? The evaluation results of various experts across different subjects in the benchmark shed light on the answer to the problem.

We analyzed that the key to the problem lies in obtaining reasonable labels for query assignment, and the performance of expert models across different subjects in benchmark tests can actually serve as a type of label. Inspired by this, we propose the Bench-CoE framework, which enables experts collaboration by effectively leveraging the strengths of different experts from benchmark evaluations, as shown in Fig. 1. Bench-CoE consists of a set of specialized expert models, a router for expert assignment, and a benchmark dataset for training the router.

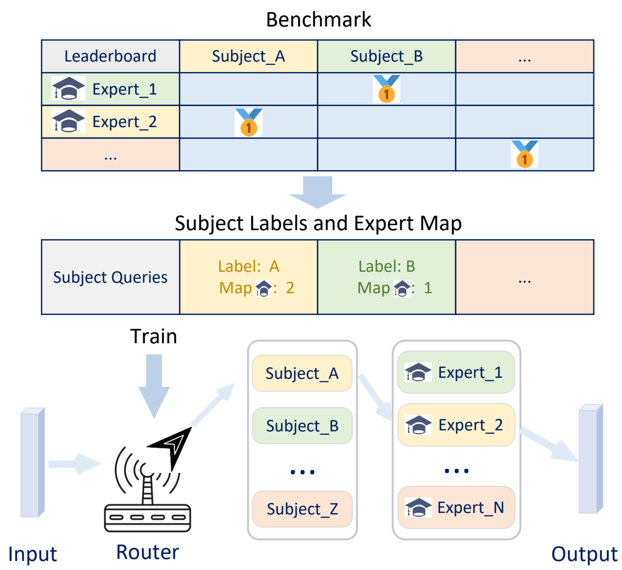

Fig. 1 The framework of Bench-CoE. It consists of a set of specialized expert models, a router for expert assignment, and a benchmark dataset for training the router.

Based on this framework, we first formulate the Query-Level Bench-CoE which is an abstraction of previous methods, such as [17, 18]. However, this approach requires re-evaluating the performance of different expert models for each query to create data labels. Although the router can effectively allocate in-distribution queries to the appropriate experts, it is expensive in terms of labels creation and struggles to generalize to tasks outside the data distribution.

To address this issue, we further propose Subject-Level Bench-CoE, which effectively exploits Subject-Level label from benchmark evaluations. The router of Subject-Level Bench-CoE includes a subject classifier and a subject-expert mapping. Firstly, the category index of the subject is used as the category label for all queries within that subject and is utilized to train the query subject classifier. Then, the subject-expert mapping is established by identifying which expert excels in a specific subject from benchmark. Defining the above subject-level labels and subject-expert mapping is effortless because some existing benchmarks [8, 9] typically provide subject labels along with evaluation results for each subject. During testing, the input query is first classified into a subject using the subject classifier. Then, based on the expert-subject mapping, the query is assigned to the expert which is most proficient in that subject for processing.

We conducted a series of experiments to evaluate the effectiveness and generalization ability of the Bench-CoE framework. The experimental results show that the query-level router performs better on in-distribution tasks, while the subject-level router demonstrates stronger generalization ability on out-of-distribution data, showcasing better adaptability and robustness. In summary, our main contributions are as follows:

- We propose a simple and efficient framework Bench-CoE for combining and assigning LLM/LMM-driven experts, which achieves flexible and efficient routing without relying on extensive labeled data and large-scale training.
- We propose Subject-Level Bench-CoE which utilizes subject-level labels from benchmark to train a subject classifier, enabling queries to be assigned to the most proficient expert based on an expert-subject mapping.
- Experiments demonstrate that our proposed CoE method outperforms individual models in multi-task scenarios, enhancing cross-domain multi-task processing performance with stronger generalization.

2 Related work

With the rapid proliferation of expert models across various domains, an increasing number of studies focus on efficiently leveraging their expertise. To this end, diverse architectural paradigms have been proposed to balance performance and efficiency. Among them, Mixture of Experts (MoE) and Collaboration of Experts (CoE) are two predominant paradigms, implementing expert routing at the token level and query level, respectively.

2.1 Mixture of Experts

MoE primarily employs a sparse activation mechanism, dynamically selecting sub-models (experts) for input query, thereby enabling large-scale parameter expansion while maintaining stable computational costs, as shown in Fig. 2 (a). The concept of MoE was first introduced by Jacobs et al., initially focusing on small-scale models and relying on a gating network to select the most suitable expert [10]. As computational resources have increased, Shazeer et al. proposed Sparsely-Gated MoE, which was the first to adopt the MoE structure in large-scale deep learning tasks and demonstrated significant performance improvements in neural machine translation [11]. However, this method still faced challenges such as imbalanced expert assignment. To address these issues, Lepikhin et al. introduced GShard, which alleviated token load balancing through auxiliary loss, random routing and expert capacity constraints [19] . Subsequently, Fedus et al. proposed Switch Transformer, which further optimized the MoE routing algorithm and successfully trained trillion-parameter MoE language models [12]. More recently, Zoph et al. introduced V-MoE, incorporating MoE mechanisms into computer vision (CV) tasks and achieving state-of-the-art results on the image classification task [20]. However, each individual expert module is part of MoE and cannot operate independently out of MoE, making it difficult to interpret the role of each expert.

2.2 Collaboration of Experts

To address these limitations, researchers have explored an alternative approach CoE which dynamically assigns query to the most suitable expert model. Current methods can be categorized based on their inference approach into two types: Multi-Inference CoE and Single-Inference CoE.

Multi-Inference CoE requires multiple models to process the input query and merge all output results, as shown in Fig. 2 (b). LLM-BLENDER [14] utilizes multiple different expert models to generate multiple candidate outputs, then employs the pair-ranker module to

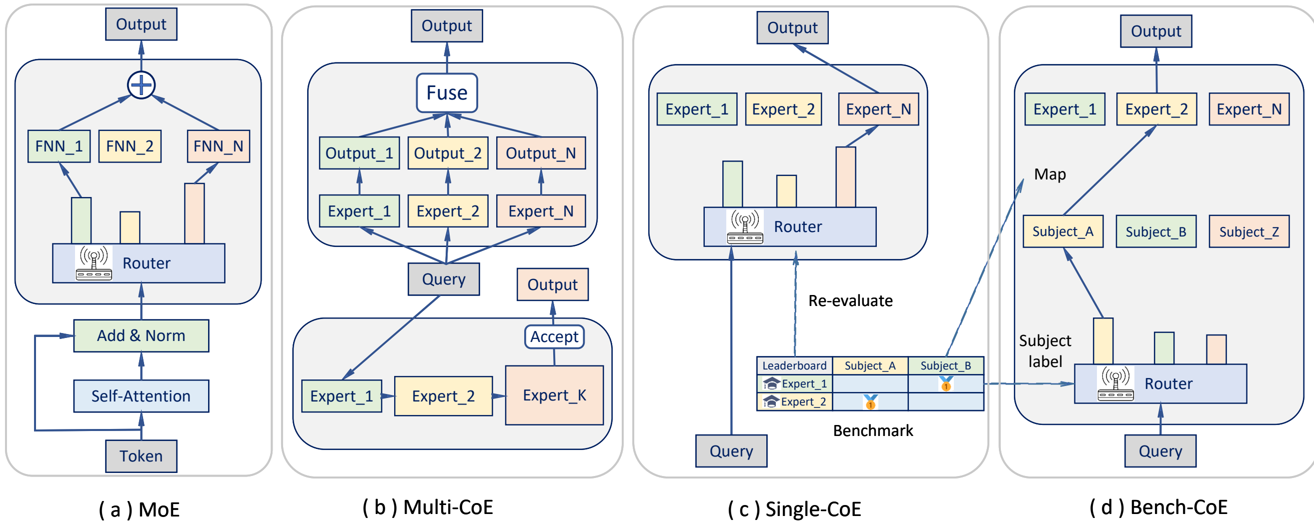

Fig. 2 (a) MoE employs multiple experts during inference and aggregates their outputs to predict the output. (b) Multi-CoE includes Parallel-CoE and Cascade-CoE, which respectively utilize multiple parallel experts to generate and integrate their individual outputs, or process sequentially through serial experts until an acceptable output is obtained. (c) The Single-CoE predicts each expert's score on a given query and selects the expert with the highest score to process the query. (d) Bench-CoE includes Query-Level and Subject-Level Bench-CoE, with the latter using benchmark subject-level labels and expert-subject mapping to assign query to the most proficient expert.

perform pairwise comparisons and select the top-ranked candidates, followed by the gen-fuser module to generate a high-quality final output. Distinct from this approach, FrugalGPT [21] explores three strategies for reducing LLM inference costs: Prompt Adaptation, LLM Approximation, and LLM Cascade. It sequentially utilizes progressively more powerful expert models to process the input task until a satisfactory response is obtained. Although this approach can achieve satisfactory results, it incurs a high computational cost as it requires multiple expert inference.

Single-Inference CoE route query to a single most suitable expert model for processing, as shown in Fig. 2 (c). This approach requires additional datasets to generate query-level routing training labels. The CoE method [13] employs a delegator mechanism combined with WGM and LGM training algorithms to improve the adaptability of expert models. Additionally, CCoE [22] utilizes a shared backbone with expert sub-networks for efficient expert integration and introduces rule-based gating mechanisms and expert planning. GraphRouter [23] and Eagle [24] leverage heterogeneous graph learning and global and local ELO ranking mechanisms, respectively, to optimize expert model selection strategies. ZOOTER [16] and TO-Router [18] improve expert selection by employing reward distillation for training routing functions and an intelligent routing mechanism to precisely match queries with expert LLMs. RouteLLM [15] and Hybrid LLM [25] reduce inference costs by utilizing dynamic model selection and query difficulty prediction with dynamic quality adjustment mechanisms, respectively. While this approach performs well within the data distribution, its generalization ability outside the distribution is suboptimal. Moreover, it requires collecting a large amount of data to create training labels for the router.

2.3 CoE with benchmark and Mapping

Some researchers have attempted to incorporate benchmarks datasets as training data during the CoE router training process, such as LLM-Bench [17]. LLM-Bench reconstructs benchmark datasets to train a router model for LLM selection and demonstrates that this problem can be transformed into a series of binary classification tasks. This approach aims to predict the performance of expert models on inputs with unknown task attributes. However, LLM-Bench has low efficiency in training the router, since it requires to re-evaluate the expert model for query labels, like the single-inference CoE as shown in Fig. 2 (c). Besides, the LLM-Bench router is not flexible for extension, since the router directly maps inputs to the expert model space. To alleviate this issue, MODULAR-CoE [26] adopts a two-step routing strategy comprising input classification and category-to-expert mapping, which achieves improved performance while reducing computational overhead. However, MODULAR-CoE requires the additional collection of a large amount of labeled data to train the router for input classification, and to train the category-to-expert mapping. Under these circumstances, the router needs to be re-trained, if there are any changes in the pool of experts.

Distinct from LLM-Bench and MODULAR-CoE, our proposed Subject-Level Bench-CoE effectively exploits subject-level labels from benchmark evaluations, without requirement to re-evaluate the queries. In addition, the routing model of our method consists of a subject classifier and a subject-expert mapping, as shown in Fig. 2 (d). Subject-expert mapping can also be directly obtained from benchmark evaluations and can be updated in real time as models evolve, without the need for retraining. The query is first classified into a subject, and then the most specialized expert is selected through this subject-expert mapping to process the query, which makes our Subject-Level Bench-CoE flexible and scalable. More importantly, our method has better generalization than LLM-Bench.

3 Framework of Bench-CoE

In this section, we first formalize the proposed Bench-CoE framework. Then, under this framework, we introduce the Query-Level Bench-CoE which is an abstraction of the ideas behind some existing methods [15–18].

3.1 Formulation of Bench-CoE

Bench-CoE consists of a set of expert models $\mathcal{M}$, a router $\mathcal{R}_\theta^N(q)$ for query q routing, and a benchmark dataset $\mathcal{B}$ for training the router. The goal of Bench-CoE is to utilize the evaluation results of expert models $\mathcal{M}$ on the benchmark dataset $\mathcal{B}$ as labels for training the router $\mathcal{R}_\theta^N(q)$, thereby enabling the reasonable assignment of input query q to the appropriate expert model for processing. The whole process can be formulated as:

$$o = f(\mathcal{M}, \mathcal{R}, q). \tag{1}$$

Where o is the final output of Bench-CoE. f represents the operation of the router $\mathcal{R}$ assigning an expert model of $\mathcal{M}$ to process the input query q.

$$\mathcal{M} = \{M_1, M_2, \dots, M_N\}. \tag{2}$$

Where N is the number of experts model, M_n represents the n-th expert model which can be used to process query q, denoted as $M_n : q \to M_n(q)$.

The router $\mathcal{R}_\theta^N(q)$ contains learnable parameters θ, and it is applied to allocate a query q to an appropriate expert model for processing. It can be a NLP model or a multi-modal, etc.

$$\mathcal{R}_\theta^N(q) : q \to \{p_1, p_2, \dots, p_N\}. \tag{3}$$

Where p_n represents the probability of routing the query q to expert model M_n for processing.

The benchmark dataset $\mathcal{B}$ may consist of K subjects and can be expressed as

$$\mathcal{B} = \{S_1, S_2, \dots, S_K\}. \tag{4}$$

Where S_k represents the k-th subject in the benchmark dataset $\mathcal{B}$. $|S_k|$ represents the number of queries in S_k, i.e. $S_k = \{q_k^1, q_k^2, \dots, q_k^{|S_k|}\}$. For example, in the MMMU benchmark validation dataset, the Math subject consists of 30 mathematical choice questions. Based on this framework, we first formalize the Query-Level Bench-CoE.

3.2 Query-Level Bench-CoE

Query-Level Bench-CoE requires evaluating the performance P of different expert models on each query in the benchmark dataset $\mathcal{B}$. Here, $P(M_n(q_k^i), t_k^i)$ represents the performance metric that measures the similarity between the output of the expert model M_n and the ground truth label t_k^i for the i-th query sample q_k^i in the k-th subject S_k of the benchmark dataset $\mathcal{B}$. For example, in the MMMU benchmark dataset, a Math subject choice question is evaluated by checking whether the option selected by the expert model matches the standard answer.

- **Expert Models Selection**

Theoretically, to achieve the best combination results, the performance of a sufficiently large number of models should be tested on each query. However, this is highly costly in applications and makes it difficult to train an efficient router. We can directly choose these specialized expert models for combination since the expert models selected based on subject specialization from the benchmark are already the best within their respective fields. Experimental results show that this simplified approach is effective.

Once the performance of different expert models on each query in the benchmark dataset is obtained, the expert model with the best performance is used as the routing label for training the router in Query-Level Bench-CoE. The query-level routing label t_k^i for query q_k^i is determined as follows:

$$t_k^i = \arg\max_l P(M_l(q_k^i), t_k^i). \tag{5}$$

If multiple models answer the same query correctly, we assign the label to the model with the highest overall score in the corresponding subject.

- **Query-Level Bench-CoE Inference.**

Given an input query q, the router $\mathcal{R}_\theta^N(q)$ of Query-Level Bench-CoE first predicts the most suitable expert model for handling the query:

$$n = \arg\max \mathcal{R}_\theta^N(q). \tag{6}$$

Where n represents the index of the most suitable expert model for handling query q. Then, the query is fed into the selected expert model M_n, producing the final output of Bench-CoE:

$$o_n^q = M_n(q). \tag{7}$$

Where o_n^q represents the output result of the expert model M_n when processing the query q.

The key to this approach is pre-evaluating different expert models on each query to obtain the benchmark data labels needed for training the router. Using benchmark datasets to generate labeled data eliminates the labor-intensive process of manual annotation required for collecting additionally methods [15, 16, 18]. As benchmarks continue to expand, the advantages of this approach will become increasingly apparent. Query routing is more accurate within the same data distribution. However, for queries out of the training data distribution, further exploration is warranted. Furthermore, some benchmarks [8, 9] only provide evaluation results at the subject level rather than the query level, making it necessary to re-evaluate expert models on the benchmark datasets to obtain training data labels.

This prompts us to consider whether there exists a routing label determination method that ensures both better generalization and without computational cost. This leads to our proposed approach: Subject-Level Bench-CoE.

4 Subject-Level Bench-CoE

We analyze that the key to solve this challenge lies in how to reasonably determine the routing labels for queries. In benchmark tests, the performance of different expert models across different subjects can

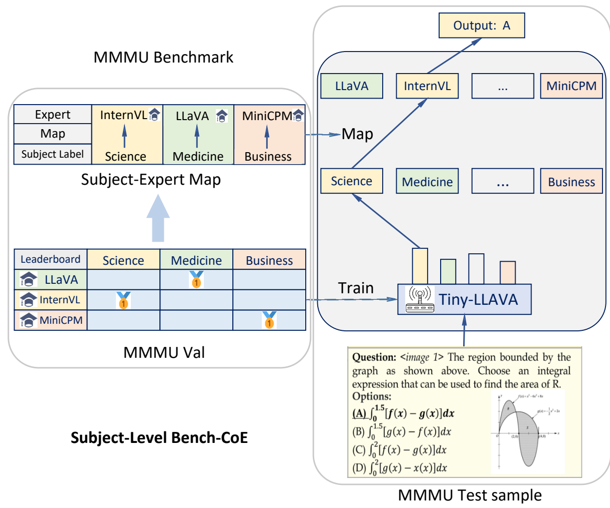

Fig. 3 Subject-Level Bench-CoE first classifies the query into a corresponding subject and then leverages the subject-expert mapping derived from benchmark to facilitate the selection of the expert most proficient in handling subject task.

actually serve as a type of label. Subject-Level Bench-CoE is based on subject-level routing labels, and an example of Subject-Level Bench-CoE is shown in Fig. 3.

The router of Subject-Level Bench-CoE consists of a subject classifier and a subject-expert mapping. The subject classifier label for the query within a subject in the benchmark dataset are assigned to that subject and some benchmarks [8,9] typically include the subject information of queries automatically. The subject-expert mapping establishes a correspondence between each subject in the benchmark and the expert model that performs best in that subject. The relationship between the expert model index n and the subject index k is established by the mapping:

$$n = \arg\max_{l} \frac{1}{|S_k|} \sum_{i=1}^{|S_k|} P(M_l(q_k^i), t_k^i). \tag{8}$$

The retrieval of subject-expert Map almost no additional effort, as the benchmark leaderboard already provides the performance results of expert models in each subject, such as [8,9]. The Map has already been completed when the leaderboard was published, meaning we only need to select the model with the highest performance in each subject.

$$Map : \{S_1, \ldots, S_K\} \to \{M_1, \ldots, M_N\}. \tag{9}$$

$K \geq N$ indicates that a single expert model may achieve the best performance across multiple subjects. Additionally, as more expert models are developed, the models that excel in specific subjects may change over time. To avoid retraining the router, we set the number of classification neurons in the last layer of the router to be equal to the number of subject categories. Then, by adjusting the mapping between subjects and expert models, we accommodate changes in subject-specialized expert models. This simple technique enhances the flexibility of the router, allowing it to adapt dynamically to changes in expert models' subject specializations without the need for retraining.

- **Subject-Level Bench-CoE Inference.**

Given an input query q, the classifier C_θ^K of Subject-Level Bench-CoE first predicts the category of the most relevant subject.

$$k = \arg\max C_\theta^K(q). \tag{10}$$

Where k represents the category of the most relevant subject for the input query q. Then, the mapping between subjects and their corresponding specialized expert models is used to determine the index of the most suitable expert model for processing the input.

$$M_n = Map(S_k). \tag{11}$$

Finally, the query is fed into the selected expert model M_n, producing the final output of Subject-Leve Bench-CoE:

$$o_n^q = M_n(q). \tag{12}$$

Where o_n^q represents the output result of the expert model M_n when processing the query q. From the above, it can be seen that our classifier C_θ^N and mapping Map together form the subject-level router $\mathcal{R}_\theta^N$.

Subject-Level Bench-CoE is more likely to correctly assign queries to the appropriate expert model compared to the Query-Level Bench-CoE. This is because, as long as the trained router can correctly classify the query into the appropriate subject category and then allocate it to the expert model proficient in that subject, it is more likely to be solved correctly compared to a non-specialized expert model. For instance, assigning a question about how to travel to Mars to a physics expert is more likely to yield a correct answer than assigning it to a literature expert.

5 Experiments

We conducted extensive experiments on both multimodal and language tasks to validate the effectiveness of our proposed method. The evaluations compare Bench-CoE with individual expert models under in-distribution, out-of-distribution, and cross-domain transfer settings.

5.1 Experimental Setup

- **Multimodal Models.**

For multimodal experiments, we use TinyLLaVA-Phi-2-SigLIP-3.1B [27] and BERT [28] as the subject classifiers for multimodal and textual inputs, respectively. TinyLLaVA combines Phi-2 with a SigLIP image encoder, and we add a linear classification layer for routing. The expert pool contains MiniCPM-V-2.6 [29], InternVL2-8B [30], and LLaVA-OV-7B [31]. These models provide complementary strengths across visual question answering, multimodal reasoning, and joint visual-text understanding, making them suitable for evaluating expert routing.

- **Evaluation Configuration.**

Query-Level and Subject-Level Bench-CoE are evaluated with the same candidate expert pool and benchmark splits in each setting. This controlled configuration isolates the influence of the routing labels and assignment mechanism from differences in the underlying expert models. At inference time, each router activates one expert for every query, and all comparisons report the accuracy of the final routed response. The resulting setup therefore supports a direct comparison of routing quality across in-distribution, out-of-distribution, and cross-domain evaluations.

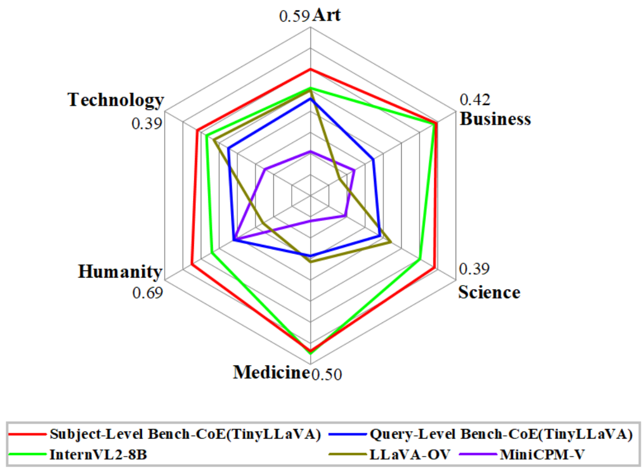

Fig. 4 Performance of the in-distribution evaluation on MMMU. The Bench-CoE routers are trained on the validation split and evaluated on the separate test split.

Table 1 In-distribution Evaluation on MMMU.

Model	Accuracy
MiniCPM-V-2.6 [29]	39.40%
InternVL2-8B [30]	44.20%
LLaVA-OV-7B [31]	41.30%
Query-Bert-Bench-CoE	41.40%
Query-TinyLLaVA-Bench-CoE	41.30%
Subject-Bert-Bench-CoE	44.50%
Subject-TinyLLaVA-Bench-CoE	**44.70%**

Table 1 shows that the two subject-level routers reach 44.50% and 44.70% accuracy, both exceeding the strongest listed individual expert at 44.20%. By contrast, the query-level variants remain near 41%, highlighting the benefit of subject-aware routing within the same MMMU evaluation split. TinyLLaVA gives the highest subject-level score, while the BERT router remains close behind. Their narrow gap indicates that the subject-based routing formulation contributes consistently across the two classifier choices.

- **Multimodal Datasets.**

MMMU [9] is a multi-discipline benchmark for multimodal understanding and reasoning across diverse academic subjects. We use its validation and test splits for the in-distribution evaluation. MMStar [32] contains multimodal questions with a distribution different from MMMU and is used to assess out-of-distribution generalization. Together, the two benchmarks allow us to evaluate both within-benchmark transfer and transfer across multimodal benchmarks.

MMMU supplies the subject structure used to construct the routing labels, whereas MMStar is reserved as a shifted benchmark for measuring transfer. Keeping the two roles separate prevents MMStar examples from contributing to router construction and yields a clean out-of-distribution assessment.

- **Language Models.**

For language experiments, BERT [28] serves as the subject classifier. The expert pool consists of Qwen2-7B-Instruct [33], Gemma-2-9B-it [34], Mathstral-7B-v0.1 [35], and Llama-3-Smaug-8B [36]. The experts exhibit complementary strengths across general instruction following, mathematical reasoning, and multi-domain language understanding.

- **Language Datasets.**

MMLU-Pro [8] extends MMLU with more challenging professional-level questions spanning a broad range of subjects. Big-Bench-Hard (BBH) [37] contains difficult language tasks that emphasize reasoning and problem solving. Their distinct task distributions and discipline-specific structure make them appropriate for cross-domain evaluation of the router.

5.2 Evaluation Protocols

- **In-distribution Evaluation.**

The Bench-CoE routers are trained with labels constructed from the MMMU validation split and evaluated on the separate MMMU test split. This protocol keeps the training and evaluation samples disjoint while preserving the same benchmark distribution.

- **Out-of-distribution Evaluation.**

The multimodal routers are trained on the MMMU validation split and evaluated on the MMStar validation split. This protocol measures whether the routing strategies transfer to a different multimodal benchmark and provides a stronger test of generalization than the in-distribution setting.

- **Cross-domain Evaluation.**

We extend Bench-CoE from multimodal tasks to language tasks by training the router on MMLU-Pro and evaluating it on BBH. Both datasets provide discipline-related structure, but they differ substantially in task composition; therefore, this protocol evaluates cross-domain transfer at both the query and subject levels.

BBH examples are not used when constructing the router from MMLU-Pro. Accuracy therefore measures whether the learned subject categories and expert assignments remain effective after the benchmark and task distributions change.

5.3 In-distribution Evaluation

We first evaluate whether the routers generalize to unseen samples from the same benchmark distribution. Following the in-distribution protocol in Section 5.2, the routers are trained on the MMMU validation split and evaluated on its separate test split. The results are shown in Table 1 and Fig. 4. This evaluation therefore measures transfer to unseen queries without introducing a benchmark-level distribution shift. All routing variants use the same candidate expert pool, keeping the comparison focused on the routing strategy.

The results indicate that Query-Level Bench-CoE is 2.80% and 2.90% less accurate than its respective expert models on MMMU. In contrast, Subject-Level Bench-CoE maintains strong generalization, achieving 0.3% and 0.5% higher accuracy than its corresponding experts. This improvement reflects the stronger generalization of a router trained with reusable subject-level labels rather than query-level labels. The consistent gain with both subject classifiers also shows that the improvement is not tied to a single router backbone. In both cases, subject-level labels preserve the expert specialization signal on previously unseen test queries.

Table 2 Performance of the Out-of-distribution Evaluation on MMMU and MMStar val.

Model	Accuracy
MiniCPM-V-2.6 [29]	54.33%
InternVL2-8B [30]	59.22%
LLaVA-OV-7B [31]	55.86%
Query-Bert-Bench-CoE	56.00%
Query-TinyLLaVA-Bench-CoE	55.62%
Subject-Bert-Bench-CoE	**60.09%**
Subject-TinyLLaVA-Bench-CoE	60.08%

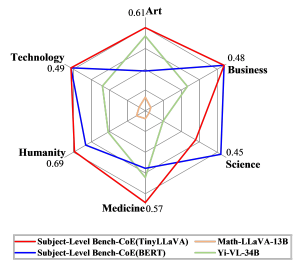

Fig. 5 Performance of the expert models and Bench-CoE across subjects on MMStar in the out-of-distribution evaluation.

5.4 Out-of-distribution Evaluation.

We next evaluate whether the multimodal routers transfer across benchmarks. The routers are trained on the MMMU validation split and evaluated on the MMStar validation split, following the out-of-distribution protocol in Section 5.2. The results are shown in Table 2 and Fig. 6.

Query-Level Bench-CoE still suffers from severe generalization deficiencies, with response accuracy 3.22% and 3.6% lower than that of its respective expert models. In contrast, Subject-Level Bench-CoE achieves 0.87% and 0.86% higher accuracy than its corresponding experts even when the evaluation distribution differs from the training benchmark, further validating the stronger generalization of subject-level routing.

Table 3 Performance of Bench-CoE and large-scale LMMs on the validation split of MMMU.

Model	Accuracy
Math-LLaVA-13B	38.3%
Yi-VL-34B	45.9%
Subject-Level Bench-CoE (BERT)	50.78%
Subject-Level Bench-CoE (TinyLLaVA)	**52.00%**

Table 3 shows that subject-level routing reaches 50.78% with BERT and 52.00% with TinyLLaVA, exceeding both large-model baselines. The result indicates that expert specialization and routing quality can compensate for model scale.

Figure 5 provides a subject-wise breakdown on MMStar. The results show how the expert models and routing strategies vary across subjects and further illustrate the benefit of selecting subject-specialized experts. The comparison in Fig. 6 likewise shows that the subject-specialized router remains competitive when the evaluation distribution differs from the training benchmark.

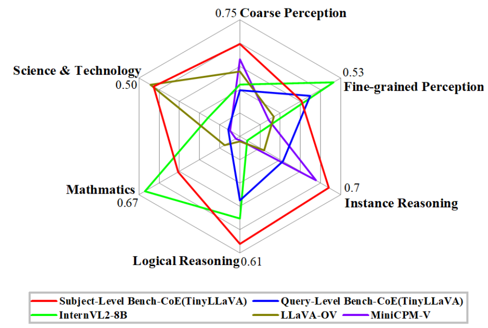

Fig. 6 Performance of the out-of-distribution evaluation on MMStar.

5.5 Comparison with Large-Scale LMMs

To compare Bench-CoE with larger standalone models, we select models with more than 10 billion parameters from the MMMU leaderboard. As shown in Table 3, Subject-Level Bench-CoE achieves 50.78% with BERT and 52.00% with TinyLLaVA, outperforming Math-LLaVA-13B and Yi-VL-34B. Routing among complementary smaller experts can therefore surpass substantially larger individual models while activating only one expert for each query.

5.6 Cross-domain Evaluation

Finally, we evaluate whether Bench-CoE transfers from multimodal routing to the language setting. The router is trained on MMLU-Pro and evaluated on BBH using the language expert pool defined in Section 5.1. The results are reported in Table 4 and Fig. 7 on the following page.

Bench-CoE remains effective for LLMs in this cross-domain setting: Query-Level and Subject-Level Bench-CoE exceed their corresponding expert models by 0.72% and 3.56%, respectively. The particularly clear gain of Subject-Level Bench-CoE further validates the transferability and generalization of subject-level expert routing.

The cross-domain results in Fig. 7 and Table 4 evaluate transfer from multimodal routing to the language setting. Bench-CoE continues to improve over the corresponding individual language experts on BBH while preserving the single-expert inference pattern used in the multimodal experiments. This controlled design keeps the inference budget comparable across the two evaluation settings. It also ensures that the reported gain comes from improved expert assignment rather than from activating additional models at inference time. The result therefore complements the multimodal evaluations with a direct test of routing transfer.

Together, these results highlight the generalization of subject-level expert routing when the framework is transferred from multimodal benchmarks to language tasks. The consistent gains across settings show that benchmark-derived subject labels remain useful beyond the distribution used to construct the router and can support expert assignment across task modalities. In particular, the larger improvement of Subject-Level Bench-CoE shows that reusable subject categories provide a stable basis for selecting complementary experts. This behavior is consistent with the out-of-distribution results observed on MMStar.

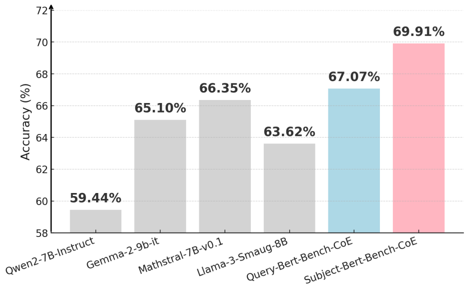

Fig. 7 Performance of the cross-domain evaluation on MMLU-Pro val and Big-Bench-Hard.

Table 4 Performance of the Cross-domain Evaluation on MMLU-Pro and BBH.

Model	Accuracy
Qwen2-7B-Instruct [33]	59.44%
Gemma-2-9b-it [34]	65.10%
Mathstral-7B-v0.1 [35]	66.35%
Llama-3-Smaug-8B [36]	63.62%
Query-Bert-Bench-CoE	67.07%
Subject-Bert-Bench-CoE	**69.91%**

6 Limitations and Future Work

Although our proposed Bench-CoE demonstrates favorable performance in terms of generalization, flexibility, and avoiding the need for data annotation, it still has the following limitation: Diversification of Model Capabilities. Previous experiments have demonstrated that CoE models achieve optimal effectiveness only when each expert model exhibits distinct specialized capabilities. The presence of a minority of models that hold significant leads across all capabilities will diminish the effectiveness enhancement of such collaborative systems. In future research, we plan to explore more balanced expert selection strategies and adaptive mechanisms to mitigate the impact of dominant experts, thereby enhancing the robustness and scalability of Bench-CoE in broader multi-task scenarios.

7 Conclusion

We propose the Bench-CoE framework, designed to facilitate effective collaboration between experts in LMMs and LLMs by leveraging benchmark evaluation results. The framework consists of the Query-Level Bench-CoE, abstracted from existing methodologies, and the Subject-Level Bench-CoE, which we introduce to enhance both the performance and scalability of the framework. Extensive experiments on multimodal and language tasks have validated the feasibility of our framework and demonstrated the generalization ability and efficiency of the Subject-Level Bench-CoE. We hope that Bench-CoE can offer a paradigm for integrating diverse experts, enhancing adaptability while reducing reliance on labeled data. It is expected to inspire the development of more robust and scalable expert collaboration systems and their application in real-world multi-task scenarios.

Ethics Statement

This research adheres to applicable ethical standards and research-integrity requirements. Our work does not involve human subjects, nor does it use datasets containing sensitive, private, or discriminatory content. All datasets employed are publicly available and used in compliance with their licenses and release practices. The methods and conclusions of this research do not contain potential malicious applications, and we have carefully evaluated and avoided possible negative societal impacts. All experiments and results comply with research integrity and academic standards.

Reproducibility Statement

We have made every effort to ensure the reproducibility of our results. The paper includes detailed descriptions of the model architecture, algorithmic procedures, and experimental settings. All datasets used are publicly available, and the preprocessing steps are documented in the supplementary materials. We have released our complete source code and experimental configurations, allowing other researchers to independently reproduce our experiments and main findings.

LLM Usage Statement

A large language model (LLM) was employed solely for grammatical error checking during the preparation of this manuscript. The LLM was not used for generating research ideas, designing experiments, analyzing results, or writing substantive scientific content. All methodological and experimental contributions are the authors' own work.

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Competing interests

The authors declare that they have no competing interests or financial conflicts to disclose.

A Scalability of Bench CoE

In Bench-CoE, particularly in the subject-level Bench-CoE, we leverage the best-performing LLM for each domain as the routing target. By directing as many questions as possible within a given subject to the "best" expert for inference, we enhance the overall accuracy of the model. However, with the rapid evolution of large language models, accompanied by the introduction of new datasets, novel models, and updated evaluation methods, the leaderboard rankings of LLMs change frequently. Under such circumstances, a fixed routing strategy in the CoE model cannot accommodate newly emerging models or adapt to shifting data distributions.

To address this limitation and improve the scalability of Bench-CoE, we designed a subject-expert mapping mechanism. Instead of directly routing inputs to a fixed best-performing model in a domain, we first classify the given input into a specific subject type. Then we leverage the subject-expert mapping to route the input to the most suitable model for that domain. This approach significantly enhances the scalability of Bench-CoE, allowing it to flexibly adapt to rapidly evolving expert models advancements by dynamically adjusting the mapping and updating routing rules.

B Scenarios Unsuitable for CoE

- **Lack of Diversification in Model Capabilities**

In our experiments with the Bench-CoE model, we selected a wide range of LLMs as candidate models and conducted extensive testing. Through these tests, we identified a common challenge in the CoE field: the issue of expert capability diversity. Specifically, this problem arises when a candidate expert lacks capability diversity on the given dataset — either significantly outperforming or underperforming all other candidate expert. Such cases negatively impact the overall performance of the CoE models, as the router is forced to route all queries either exclusively to or completely away from this model to achieve optimal results. This creates a significant challenge for training the router.

Looking ahead, we believe this issue can be mitigated with the development of dynamic routing strategies and adaptive candidate LLM selection mechanisms. These advancements will enable the CoE model to better handle capability imbalances among candidate LLMs, paving the way for more robust and flexible routing solutions.

References

[1] Radford A, Wu J, Child R, Luan D, Amodei D, Sutskever I, others . Language models are unsupervised multitask learners. OpenAI blog, 2019, 1(8): 9

[2] Brown T B, Mann B, Ryder N, Subbiah M, Kaplan J, Dhariwal P, Neelakantan A, Shyam P, Sastry G, Askell A, Agarwal S, Herbert-Voss A, Krueger G, Henighan T, Child R, Ramesh A, Ziegler D M, Wu J, Winter C, Hesse C, Chen M, Sigler E, Litwin M, Gray S, Chess B, Clark J, Berner C, McCandlish S, Radford A, Sutskever I, Amodei D. Language models are few-shot learners. In: Larochelle H, Ranzato M, Hadsell R, Balcan M, Lin H, eds, Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual. 2020

[3] Liu H, Li C, Wu Q, Lee Y J. Visual instruction tuning. In: NeurIPS. 2023

[4] Zhu D, Chen J, Shen X, Li X, Elhoseiny M. Minigpt-4: Enhancing vision-language understanding with advanced large language models. In: ICLR. 2024

[5] Tjong Kim Sang E F. Introduction to the CoNLL-2002 shared task: Language-independent named entity recognition. In: COLING-02: The 6th Conference on Natural Language Learning 2002 (CoNLL-2002). 2002

[6] Bowman S R, Angeli G, Potts C, Manning C D. A large annotated corpus for learning natural language inference. In: Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing. 2015, 632–642

[7] Rajpurkar P, Zhang J, Lopyrev K, Liang P. SQuAD: 100,000+ questions for machine comprehension of text. In: Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing. 2016, 2383–2392

[8] Wang Y, Ma X, Zhang G, Ni Y, Chandra A, Guo S, Ren W, Arulraj A, He X, Jiang Z, others . Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. ArXiv preprint, 2024, abs/2406.01574

[9] Yue X, Ni Y, Zhang K, Zheng T, Liu R, Zhang G, Stevens S, Jiang D, Ren W, Sun Y, others . Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024, 9556–9567

[10] Jacobs R A, Jordan M I, Nowlan S J, Hinton G E. Adaptive mixtures of local experts. Neural computation, 1991, 3(1): 79–87

[11] Shazeer N, Mirhoseini A, Maziarz K, Davis A, Le Q, Hinton G, Dean J. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. arXiv preprint arXiv:1701.06538, 2017

[12] Fedus W, Zoph B, Shazeer N. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. Journal of Machine Learning Research, 2022, 23(120): 1–39

[13] Zhang Y, Chen Z, Zhong Z. Collaboration of experts: Achieving 80% top-1 accuracy on imagenet with 100m flops. In: Chaudhuri K, Jegelka S, Song L, Szepesvári C, Niu G, Sabato S, eds, International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA. 2022, 26068–26084

[14] Jiang D, Ren X, Lin B Y. Llm-blender: Ensembling large language models with pairwise ranking and generative fusion. In: ACL (1). 2023, 14165–14178

[15] Ong I, Almahairi A, Wu V, Chiang W L, Wu T, Gonzalez J E, Kadous M W, Stoica I. Routellm: Learning to route llms with preference data. ArXiv preprint, 2024, abs/2406.18665

[16] Lu K, Yuan H, Lin R, Lin J, Yuan Z, Zhou C, Zhou J. Routing to the expert: Efficient reward-guided ensemble of large language models. In: NAACL-HLT. 2024, 1964–1974

[17] Shnitzer T, Ou A, Silva M, Soule K, Sun Y, Solomon J, Thompson N, Yurochkin M. Large language model routing with benchmark datasets. arXiv preprint arXiv:2309.15789, 2023

[18] Stripelis D, Hu Z, Zhang J, Xu Z, Shah A, Jin H, Yao Y, Avestimehr S, He C. Polyrouter: A multi-llm querying system. arXiv e-prints, 2024, arXiv–2408

[19] Lepikhin D, Lee H, Xu Y, Chen D, Firat O, Huang Y, Krikun M, Shazeer N, Chen Z. Gshard: Scaling giant models with conditional computation and automatic sharding. arXiv preprint arXiv:2006.16668, 2020

[20] Riquelme C, Puigcerver J, Mustafa B, Neumann M, Jenatton R, Susano Pinto A, Keysers D, Houlsby N. Scaling vision with sparse mixture of experts. Advances in Neural Information Processing Systems, 2021, 34: 8583–8595

[21] Chen L, Zaharia M, Zou J. Frugalgpt: How to use large language models while reducing cost and improving performance. ArXiv preprint, 2023, abs/2305.05176

[22] Huang S, Pan J, Zheng H. Ccoe: A compact llm with collaboration of experts. ArXiv preprint, 2024, abs/2407.11686

[23] Feng T, Shen Y, You J. Graphrouter: A graph-based router for llm selections. ArXiv preprint, 2024, abs/2410.03834

[24] Zhao Z, Jin S, Mao Z M. Eagle: Efficient training-free router for multi-llm inference. ArXiv preprint, 2024, abs/2409.15518

[25] Ding D, Mallick A, Wang C, Sim R, Mukherjee S, Rühle V, Lakshmanan L V S, Awadallah A H. Hybrid LLM: cost-efficient and quality-aware query routing. In: ICLR. 2024

[26] Jain S, Raju R, Li B, Csaki Z, Li J, Liang K, Feng G, Thakkar U, Sampat A, Prabhakar R, others . Composition of experts: A modular compound ai system leveraging large language models. arXiv preprint arXiv:2412.01868, 2024

[27] Zhou B, Hu Y, Weng X, Jia J, Luo J, Liu X, Wu J, Huang L. Tinyllava: A framework of small-scale large multimodal models. arXiv preprint arXiv:2402.14289, 2024

[28] Devlin J, Chang M W, Lee K, Toutanova K. Bert: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers). 2019, 4171–4186

[29] Yao Y, Yu T, Zhang A, Wang C, Cui J, Zhu H, Cai T, Li H, Zhao W, He Z, others . Minicpm-v: A gpt-4v level mllm on your phone. ArXiv preprint, 2024, abs/2408.01800

[30] Chen Z, Wang W, Tian H, Ye S, Gao Z, Cui E, Tong W, Hu K, Luo J, Ma Z, others . How far are we to gpt-4v? closing the gap to

commercial multimodal models with open-source suites. ArXiv preprint, 2024, abs/2404.16821

[31] Li B, Zhang Y, Guo D, Zhang R, Li F, Zhang H, Zhang K, Li Y, Liu Z, Li C. Llava-onevision: Easy visual task transfer. ArXiv preprint, 2024, abs/2408.03326

[32] Chen L, Li J, Dong X, Zhang P, Zang Y, Chen Z, Duan H, Wang J, Qiao Y, Lin D, others . Are we on the right way for evaluating large vision-language models? ArXiv preprint, 2024, abs/2403.20330

[33] Yang A, Yang B, Hui B, Zheng B, Yu B, Zhou C, Li C, Li C, Liu D, Huang F, others . Qwen2 technical report. ArXiv preprint, 2024, abs/2407.10671

[34] Team G. Gemma. Kaggle model card, 2024

[35] Jiang A Q, Sablayrolles A, Mensch A, Bamford C, Chaplot D S, Casas D d l, Bressand F, Lengyel G, Lample G, Saulnier L, others . Mistral 7b. ArXiv preprint, 2023, abs/2310.06825

[36] Meta A. Introducing meta llama 3: The most capable openly available llm to date. Meta AI, 2024

[37] Suzgun M, Scales N, Schärli N, Gehrmann S, Tay Y, Chung H W, Chowdhery A, Le Q V, Chi E H, Zhou D, Wei J. Challenging big-bench tasks and whether chain-of-thought can solve them. In: ACL (Findings). 2023, 13003–13051